

NEXO 360 - Project Timeline

Completion deadline: July 10, 2026 | Planning window: June 12 - July 10, 2026 | MVP-focused competition delivery

This timeline organizes the execution of NEXO 360 into practical milestones for a cross-platform Flutter application with Firebase backend services and a public website for event information and external registrations. The plan is designed for an MVP demo by July 10, prioritizing role-based access, core workflows, QR permission validation, event registration, and testing over a full production rollout.

Planning assumptions and major dependencies

Assumption	Details
Delivery strategy	MVP first: deliver a stable demo with core workflows before adding advanced features such as AI assistance, full analytics, or complete production hardening.
Technical stack	Flutter for the cross-platform app; Firebase Authentication, Firestore, Storage, and optional Hosting for backend and website support.
Critical dependency	Role and permission rules must be defined before security rules, screen access, and module workflows can be completed.
Risk control	Feature scope is intentionally limited to a competition-ready MVP by July 10; extra features should remain in a stretch backlog.

Phase summary

Phase	Start	End	Primary deliverable
0. Initiation & Scope	Jun 12	Jun 13	Approved MVP scope and prioritized backlog
1. User Structure & Access Control	Jun 14	Jun 18	Role-based login, user profile model, and test users
2. Design Phase	Jun 18	Jun 24	Clickable prototype for app and public website
3. Core Module Development	Jun 24	Jul 06	Integrated School Portal, QR permissions, Youth Movement, and dashboard MVP
4. Testing & Validation	Jul 06	Jul 10	Tested, polished, submission-ready demo package

Detailed milestone timeline

ID	Date range	Phase	Milestone	Deliverable / checkpoint	Dependencies
M01	Jun 12	0. Initiation & Scope	Kickoff and MVP scope lock Confirm project objective, competition deadline, target platforms, MVP modules, and acceptance criteria.	Approved MVP scope, project assumptions, and success criteria. Checkpoint: Scope approved	None
M02	Jun 12 - Jun 13	0. Initiation & Scope	Feature prioritization and release plan Separate must-have competition demo features from phase-two features to keep delivery realistic.	Prioritized backlog: MVP, stretch, and future features. Checkpoint: MVP feature list approved	M01
M03	Jun 14 - Jun 15	1. User Structure & Access Control	Role and permission matrix Define account types and role-based access for technical users, teachers, students, organizers, committee members, and guests.	Role-permission matrix and access rules per module. Checkpoint: Roles validated against scope	M02
M04	Jun 15 - Jun 16	1. User Structure & Access Control	Firebase project and data environments Create Firebase project, configure Authentication, Firestore, Storage, and development/testing collections.	Firebase environment ready for development. Checkpoint: Firebase services enabled	M03
M05	Jun 16 - Jun 18	1. User Structure & Access Control	Authentication and user profile model Implement login structure, user profile fields, unique user codes, and test accounts for each role.	Working login prototype with seeded users and role-specific home access. Checkpoint: Role-based login demo	M04
M06	Jun 18 - Jun 19	2. Design Phase	Information architecture and navigation map Define the main menu structure, module navigation, and user journeys for each role.	App map and navigation flow. Checkpoint: Navigation approved	M03
M07	Jun 19 - Jun 21	2. Design Phase	Application wireframes Design low-to-mid fidelity screens for login, dashboard, school portal, QR permissions, Youth Movement, and settings.	Wireframe set for main application screens. Checkpoint: Wireframes reviewed	M06
M08	Jun 20 - Jun 22	2. Design Phase	Public website design Design event information pages, registration form, document upload flow, and confirmation states.	Public website layout and form flow. Checkpoint: Website flow approved	M06
M09	Jun 22 - Jun 24	2. Design Phase	Visual style guide and prototype review Apply NEXO 360 branding, colors, typography, iconography, and create a clickable prototype for review.	Clickable prototype and visual guide. Checkpoint: Design freeze for MVP	M07, M08
M10	Jun 24 - Jun 25	3. Core Module Development	Flutter project structure and routing Set up Flutter app architecture, responsive layout strategy, routes, reusable components, and platform targets.	Project foundation running on Android emulator and Windows build target. Checkpoint: Base app runs	M05, M09
M11	Jun 25 - Jun 26	3. Core Module Development	Firebase integration and security rules draft Connect Flutter to Firebase services, configure Firestore collections, Storage paths, and draft role-aware security rules.	App connected to Firebase with initial secure access rules. Checkpoint: Read/write tests passed	M04, M10
M12	Jun 26 - Jun 29	3. Core Module Development	School Portal - assignments, announcements, and files Implement course notices, task creation, file/link attachments, and student task visibility.	Usable School Portal MVP for teachers and students. Checkpoint: Teacher-to-student workflow verified	M11
M13	Jun 28 - Jun 30	3. Core Module Development	School Portal - attendance and grades MVP Implement simplified attendance records, grade entry placeholders, and student view of accumulated scores.	Attendance and grade viewing MVP. Checkpoint: Sample class validated	M11, M12
M14	Jun 29 - Jul 02	3. Core Module Development	QR permission creation and validation Build teacher permission form, student selection, time-bound permission records, QR generation, and scan validation logic.	End-to-end QR permission workflow. Checkpoint: Valid, expired, and unauthorized QR cases tested	M05, M11
M15	Jun 30 - Jul 03	3. Core Module Development	Youth Movement registration website Build public event pages, external registration form, PDF upload, and submission storage.	Public website registration flow. Checkpoint: Registration saved with PDF attachment	M08, M11

ID	Date range	Phase	Milestone	Deliverable / checkpoint	Dependencies
M16	Jul 02 - Jul 04	3. Core Module Development	Youth Movement admin approval and check-in Implement review, approve/reserve status, accepted list, and participant arrival check-in workflow.	Registration approval and check-in dashboard. Checkpoint: Approved registration appears in event database	M15
M17	Jul 03 - Jul 05	3. Core Module Development	Inventories, requests, and committee communication Create simplified inventory records, committee request feed, and real-time communication channels.	Committee operations MVP. Checkpoint: Request-to-response workflow tested	M11, M16
M18	Jul 04 - Jul 06	3. Core Module Development	Real-time map and event information Implement simplified live map/croquis view with editable event locations and read-only public visibility.	Live map MVP with event status updates. Checkpoint: Map updates visible to viewers	M16
M19	Jul 05 - Jul 06	3. Core Module Development	Integrated dashboard and cross-module links Connect major modules through a central dashboard showing pending tasks, active permissions, registrations, and alerts.	Unified NEXO 360 dashboard. Checkpoint: End-to-end demo path complete	M12, M14, M16, M17, M18
M20	Jul 06 - Jul 07	4. Testing & Validation	Test plan and simulated data setup Prepare test users, sample courses, event records, QR permissions, registrations, and committee scenarios.	Structured test dataset and checklist. Checkpoint: Test environment ready	M19
M21	Jul 07 - Jul 08	4. Testing & Validation	Functional and role-based testing Validate that each role can only access the correct screens and that core workflows operate as expected.	Functional test results and defect list. Checkpoint: Critical workflows pass	M20
M22	Jul 08 - Jul 09	4. Testing & Validation	QR security and communication validation Test QR permissions for validity windows, unauthorized access, expired permissions, and committee messaging efficiency.	QR validation report and communication test notes. Checkpoint: Security checks passed	M21
M23	Jul 08 - Jul 09	4. Testing & Validation	Bug fixes and polish Resolve critical defects, improve confusing screens, verify responsive layout, and polish demo content.	Stable MVP candidate. Checkpoint: No critical defects open	M21, M22
M24	Jul 10	4. Testing & Validation	Final build, documentation, and presentation readiness Prepare final build/demo package, timeline documentation, screenshots, and handoff notes for presentation.	Final MVP demo package and project documentation. Checkpoint: Submission-ready by July 10	M23

Key checkpoints

Date	Checkpoint	Acceptance criteria	Milestones
Jun 12	Scope approved	MVP scope, assumptions, and competition deliverables are confirmed.	M01-M02
Jun 18	Role-based login demo	At least one test account per role reaches the correct dashboard.	M03-M05
Jun 24	Design freeze for MVP	Core app and website screens are approved for development.	M06-M09
Jun 26	Firebase integration verified	Flutter app reads/writes sample role-protected data.	M10-M11
Jul 02	QR permission workflow complete	Valid, expired, and unauthorized QR cases behave correctly.	M14
Jul 04	Registration approval working	A website registration with PDF can be approved and sent to event records.	M15-M16
Jul 06	Integrated demo path complete	Dashboard connects School Portal, QR permissions, and Youth Movement modules.	M17-M19
Jul 09	Validation complete	Critical workflows tested and no critical defects remain open.	M20-M23
Jul 10	Submission-ready	Final MVP demo, documentation, and project timeline are ready.	M24

Dependency map

Dependency area	Dependency logic
Access control before development	M03-M05 must be ready before route guards, Firestore security rules, and role-specific module screens are finalized.
Design before UI development	M06-M09 define the user journeys and screens that M10-M19 implement.
Firebase before modules	M11 is a foundation dependency for School Portal, QR permission records, registrations, inventories, map updates, and communication.
Registration before check-in	M15 must exist before M16 can review, approve, reserve, or check in event participants.
Integrated dashboard before validation	M19 must be complete before the final test dataset and workflow validation begin.

References

Flutter: <https://flutter.dev/> | Firebase Auth: <https://firebase.google.com/docs/auth> | Cloud Firestore: <https://firebase.google.com/docs/firestore> | Cloud Storage: <https://firebase.google.com/docs/storage> | Firebase Hosting: <https://firebase.google.com/docs/hosting>